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Strain-coupled one-dimensional turbulence: a single-line closure for the rapid pressure–strain and its measured limits

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The rapid distortion of turbulence by a mean strain is central to flows from wind-tunnel contractions to the stagnation regions of lifting surfaces, yet its economical prediction founders on a closure problem: the rapid pressure–strain term is a nonlocal three-dimensional spectral integral, while affordable models evolve at most one-dimensional or single-point information. We present a strain-coupled formulation of one-dimensional turbulence (ODT) and, as its analytical core, a systematic treatment of what a single line of turbulence data determines about the rapid pressure–strain term: the obstruction to a single-line closure is stated exactly, sharp bounds on the rapid term are derived from line data alone, and the closure assumption of axisymmetry about the line is not assumed but measured—it is exact for axisymmetric distortion and its plane-strain error is quantified as a function of accumulated strain. A strained-box direct numerical simulation with an exact rapid-distortion companion provides the independent spectral benchmark: it shows that the correct linear reference for line-projected spectra is broadband rather than a rigid spectral translation, and a three-way comparison localises the model's remaining deficiency in its wavenumber-uniform rapid kinematics, quantified by a distance-from-linear-theory envelope as a function of strain rapidity. Hierarchical application of the model's energy-redistributing kernel to eddy sub-scales is shown to supply the measured missing ingredient—scale-local relaxation—where it reaches the scales of the imbalance.

Key words:

1. Introduction

[M4: introduction rewrite — see markers above.]

Abstract must not spill onto p.2

2. Strain-coupled one-dimensional turbulence

[M-formulation: transplant per markers.]

3. Verification against rapid-distortion theory

[M-verification: transplant per markers.]

4. The linear reference and the distorted spectrum

4.0.1. *Compression without cascade: the model's kinematic reference*

With the eddy events suppressed and zero molecular viscosity, the mean strain is the only process acting on the resolved field, and its effect on the line is purely kinematic. The line length contracts as $\mathcal{L}(e) = \mathcal{L}_0\, e^{A_{22}e}$, so every resolved wavenumber is rescaled as $k(e) = k(0)\, e^{-A_{22}e}$—the wavevector kinematics it shares with rapid-distortion theory (Batchelor & Proudman 1954; Hunt & Carruthers 1990)—while the continuous redistribution operator of § **??** acts uniformly on every resolved wavenumber. The combination transfers no energy between scales: a rigid translation of the line spectrum toward higher wavenumber. We emphasise at the outset that this rigid translation is a property of the strain-coupled *line*—uniform dilatation composed with a wavenumber-uniform operator—and not of linear distortion theory itself; the correct linear reference for the projected spectrum is constructed in § 4.0.2 below. As a check on the implementation, however, the rigid translation is an exact, parameter-free target.

Figure 2 confirms it. The component spectra translate toward higher k without change of shape—the peak neither broadens nor develops an inertial range—and the spectral centroid follows $e^{-A_{22}e}$ to within discretisation error, reaching 7.0 at $e = 3.9$ against the exact 7.03, with the three components migrating identically. The single-point moments are unchanged from the no-dilatation case, since a uniform rescaling of position leaves the length-weighted velocity statistics invariant. Compression therefore acts entirely on the spectrum and reproduces the model's kinematic prediction exactly. It is the kinematic reference (the dashed line in all centroid figures below) against which the full model is measured.

4.0.2. *The linear reference for the projected spectrum: exact RDT*

The rigid translation above must not be mistaken for the prediction of linear rapid-distortion theory for the quantity the line actually carries. In the full linear theory the amplitude of each Fourier mode evolves at a rate that depends on the orientation of its three-dimensional wavevector (equation **??**); the one-dimensional component spectra are projections—integrals over the transverse wavenumber plane—and there is no reason for a projection of orientation-dependent amplification to translate rigidly. Whether it does is a question of calculation, and the answer is that it does not. Using the closed-form Cauchy solution of the linear problem applied to isotropic initial fields and projected onto a line (the exact-RDT companion calculations described in § 4.1), we define the shape ratio $R_{nn}(\kappa_2)$ as the projected component spectrum at strain e, normalised by its integral, divided by the rigidly translated initial spectrum normalised likewise; $R_{nn} \equiv 1$ at every κ_2 if and only if the translation is rigid. At $e = 1$ under plane strain the upwash ratio R_{22} falls monotonically from 1.60 at half the initial spectral centroid to 0.38 at twelve times it: exact linear theory redistributes the projected upwash spectrum toward low wavenumber by a factor of four across the resolved range, because modes aligned with the line ($\kappa \to \kappa_2$) are amplified least. The transverse components are affected more weakly (R_{11} from 1.27

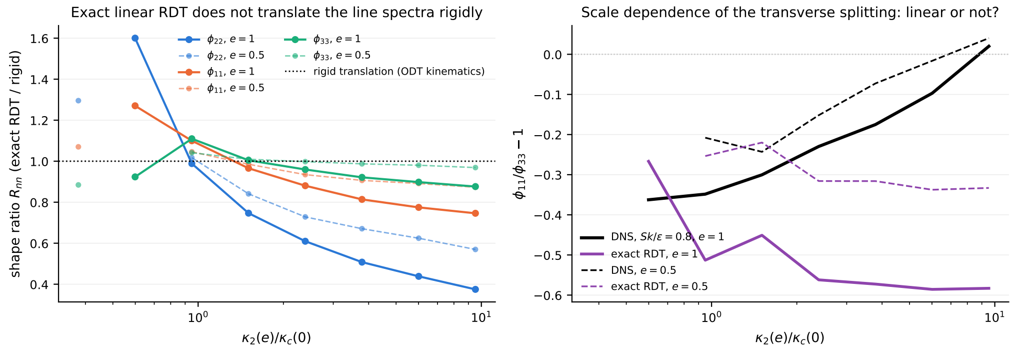

Figure 1. Exact linear theory does not translate the projected spectra rigidly. Left: shape ratio $R_{nn}(\kappa_2)$ of the exactly distorted, line-projected component spectra to a rigid translation, at $e = 1$ under plane strain (closed-form Cauchy solution, isotropic initial fields, four realisations). Right: the transverse splitting $\phi_{11}/\phi_{33} - 1$ versus wavenumber in exact linear theory and in the strained-box DNS of § 4.1.

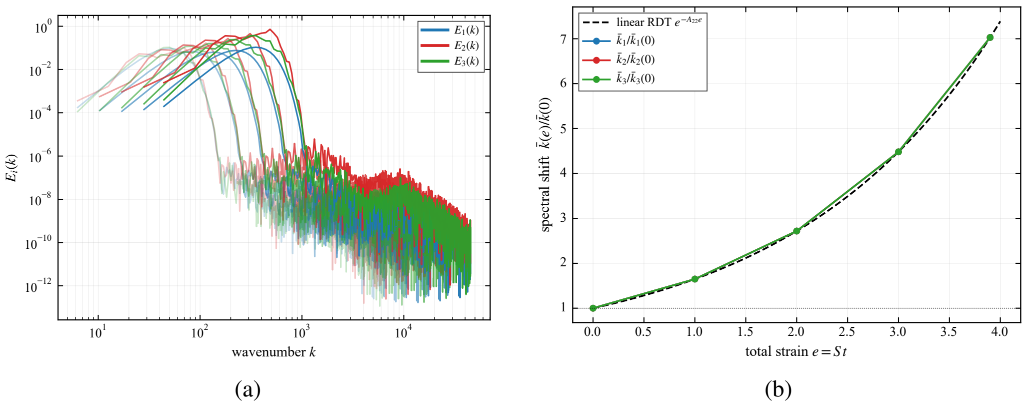

Figure 2. Compression without cascade (eddy events suppressed, inviscid). (*a*) Component spectra $E_i(k)$ at increasing total strain (lighter to darker); the spectra translate to higher k without changing shape. (*b*) Spectral-centroid migration $\bar{k}(e)/\bar{k}(0)$ for each component (symbols) against the kinematic prediction $e^{-A_{22}e}$ (dashed). The centroid follows the prediction exactly and the three components coincide, confirming purely geometric compression of the line.

to 0.75; R_{33} within ten per cent of unity), so the projected anisotropy of linear theory is itself scale-dependent. Figure 1 shows the ratios.

Two consequences run through the remainder of the paper. First, the comparison of the full model against the rigid translation (§ **??**) measures the departure of the model from its own kinematics; the departure of the *flow* from linear theory must be measured against the exact projected reference, and the two differ already at the linear level. Second, the model's rigid spectral kinematics is itself a structural approximation—linear theory deepens the projected anisotropy with wavenumber, which a wavenumber-uniform operator cannot represent—and § 4.1.1 quantifies the resulting deficiency against both the exact reference and DNS.

[M-spectrum: transplant sec:spec-bvc + compressed CMK here.]

4.1. *Validation against a strained-box DNS with an exact-RDT companion*

The comparisons above validate the model at the Reynolds-stress level. The referent quantity for the leading-edge application, however, is spectral, and the objection that

$\Delta b_{22}(\kappa_2)$, band κ_2/κ_c:	0.5–0.75	0.75–1.2	1.2–1.9	1.9–3	3–4.8	4.8–7.6	7.6–12
exact projected RDT	+0.146	+0.008	−0.020	−0.041	−0.050	−0.054	−0.056
DNS	+0.032	−0.033	−0.029	−0.032	−0.036	−0.040	−0.046
strain-coupled ODT	+0.100	+0.101	+0.114	+0.111	+0.127	+0.099	+0.147

Table 1. Strain-induced upwash anisotropy per wavenumber band at $e = 1$, $Sk_t/\varepsilon = 0.8$, each system referenced to its own unstrained state. Bands below $0.75\,\kappa_c$ rest on a few box modes and are indicative only.

the emergent-spectrum results of § 4 compare the model only with itself must be met with an independent spectral benchmark under the same strain. To that end we performed direct numerical simulations of plane-strained homogeneous turbulence in a deforming-frame (Rogallo-type) pseudo-spectral formulation: a 128^3 box, decaying isotropic precursor, then plane strain $A = S\,\mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ applied at strain-rate parameters $Sk_t/\varepsilon \in \{0.8, 16\}$ at onset, carried to total strain $e = 1$, four independent realisations ($\kappa_{\max}\eta \approx 0.76$ at $e = 1$: a pilot resolution, so small-scale statements below are qualitative). Alongside every DNS realisation runs an *exact-RDT companion*: the same initial field evolved under the closed-form Cauchy solution of the linearised equations and projected onto lines identically to the DNS and to ODT. The companion is what § 4.0.2 used to construct the linear reference; here it completes a three-way comparison in which model, linear theory and DNS are reduced to the same one-dimensional observables. On the ODT side the ensembles use an unstrained precursor: the line is evolved as ordinary ODT until the initialisation transient has relaxed and the eddy population is statistically active, and the strain is switched on only then, so that the strained evolution starts from a physically conditioned state rather than from the synthetic initial condition. All ODT statistics in this subsection are medians over 1024 realisations per case with bootstrap confidence intervals; median statistics are used because the strained band energies are strongly intermittent across realisations, and ensemble means of spectral ratios are dominated by rare events at any practical sample size.

4.1.1. *Three-way spectral anisotropy: ODT, exact RDT, DNS*

The comparison quantity is the strain-induced change of the spectral component anisotropy, $\Delta b_{nn}(\kappa_2) = \phi_{nn}/\sum_m \phi_{mm} - \frac{1}{3}$ at strain e, minus the same quantity evaluated in the system's own unstrained state with each mode mapped to its strained wavenumber. Referencing each system to its own $e = 0$ state cancels viscous decay, which is component-blind, and the longitudinal–transverse structure of the isotropic line spectra, which differs between a three-dimensional box and the component-symmetric ODT line. Figure 3 and table 1 give the result at $e = 1$, $Sk_t/\varepsilon = 0.8$, on common bands of $\kappa_2(e)/\kappa_c(0)$ with κ_c the initial spectral centroid.

The structure of the table is the finding. Exact linear theory concentrates the strain-induced upwash anisotropy at the energy-containing scales and *removes* it from the small scales, because modes aligned with the line are amplified least; the DNS shows the same shape, weakened at the large scales by the nonlinear return. The strain-coupled ODT distributes its moment-level anisotropy essentially uniformly over wavenumber—the signature, anticipated in § 4.0.2, of a wavenumber-uniform redistribution operator acting on a uniformly dilated line. The integrated anisotropy is nevertheless correct (the model tracks the closure at the moment level, § **??**), so the deficiency is one of allocation across scales, not of amount. For the leading-edge application the operative consequence is that the high-wavenumber anisotropy of the upwash spectrum, which the Amiet response weights, has

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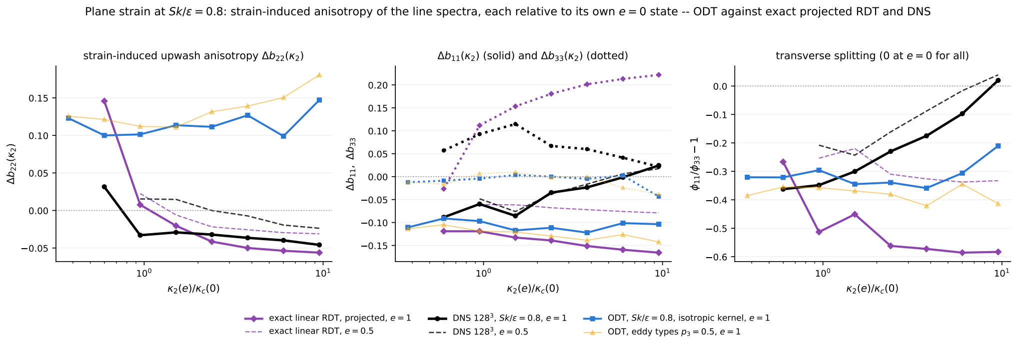

Figure 3. Three-way comparison at $Sk_t/\varepsilon = 0.8$: strain-induced anisotropy of the line spectra relative to each system's own unstrained state; strain-coupled ODT (1024 realisations) against exact projected RDT and DNS (128^3, four realisations); $e = 1$ solid, $e = 0.5$ dashed.

the wrong sign in the present model; the closure element that must carry the correction—a wavenumber-dependent rapid operator in place of the uniform one—is exactly the spectral kernel formulation of § 5, whose exercise is thereby motivated by a measured deficit rather than by completeness.

4.1.2. *Distance from linear theory as a function of strain rapidity*

The three-way comparison resolves the anisotropy by wavenumber at one strain-rate parameter. A complementary scalar measure tracks the *whole* spectral shape across strain and strain rapidity: at each total strain the median ODT component line spectra are fitted, jointly in the upwash and transverse components, by the exactly distorted spectrum of linear theory—the closed-form Cauchy distortion of an isotropic von Kármán field, projected onto the line, with only the pre-distortion amplitude and energy wavenumber free. The root-mean-square log-residual of that two-parameter fit is a scale-resolved distance between the model's strained spectra and exact linear theory. Figure 4 shows it for precursor-conditioned ensembles at $Sk_t/\varepsilon \approx 0.4$ and ≈ 8 at onset (1024 realisations each, identical precursor state: the two fits coincide at $e = 0$ at thirteen per cent, the residual shape mismatch of a relaxed ODT line against the von Kármán form).

The distance grows with strain in both cases, but *faster at rapid strain*: to approximately 22 per cent at $e = 2$ for $Sk_t/\varepsilon \approx 0.4$ against approximately 24 per cent, saturating early, at $Sk_t/\varepsilon \approx 8$, with the separation established already by $e = 0.5$. If the model's rapid response were spectrally faithful, the rapid curve is the one that should approach zero: eddy events are subdominant there by construction. That the deviation is largest precisely where the eddies matter least localises the deficiency in the rapid kinematics themselves—the wavenumber-uniform operator and the rigid dilatation—and not in the eddy or relaxation statistics, in quantitative agreement with the three-way comparison. This measured validity envelope accompanies the model into any application: predictions inherit a spectral-shape uncertainty of order the plotted distance at the operating point's strain and rapidity.

5. Closure of the rapid pressure–strain term on a single line

[M3: the Section-5 rebuild — the largest single writing task.]

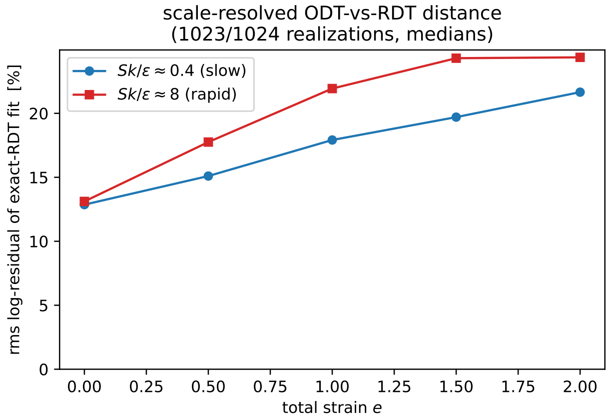

Figure 4. Scale-resolved distance between the strained ODT line spectra and exact linear theory: rms log-residual of the two-parameter exact-RDT-distorted von Kármán fit, against total strain, for slow and rapid onset strain-rate parameters (medians over 1024 realisations; identical precursor state at $e = 0$).

6. Measured limits, and scale-local relaxation

[M-limits: compressed Kerstein arc — write after Alan's next reply settles the iterate-vs-depth question, or with current results if the manuscript moves first.]

7. Consequence for leading-edge noise

[M5: detachable LE section — decision D1.]

8. Conclusions

[M4: conclusions rewrite.]

Appendix A. The forward map and the bound kernels

[Appendix A: transplant from notes/los_estimator_note.tex.]

Appendix B. Numerical parameters and initialisation

[Appendix B: numerics + IC whitening + statistics policy.]

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